

0 - 255

IPv4 32 bit
~~IPv6 64 bit~~
 IPv6 128 bit

A	0-127	no-H-K-H
B	128-191	no-no-H-H
C	192-223	no-no-no-H
D	224-239	multicast
E	240-255	Broadcast

IPv6

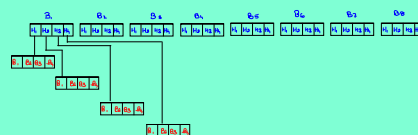
Total length of IPv6 address is 128 bit

All the 128 bits are separated into 8 blocks

One block contains 4 Hexa. Hence

1 Hexa = 4 bit

It is in Hexa decimal format



IPv6	Host
2a03:2880:f11c:8183	face:b00c:0000:25de
64	64

Compression method :-

* Removing Leading Zero in a block

* Zero block have compression

3002; Fc301; 41C7; 00a8; bcd0; 2347; 8000; 1243

Removing Leading Zero in a block

1 2 3 4

0 0 0 0 - 0000
 0 0 0 1 - 0001
 1 0 0 0 - 1000
 0 0 0 0 - 0

3002; Fc301; 41C7; 00a8; bcd0; 2347; 8000; 1243

3002; Fc30; 41C7; 08; bc30; 2347; 8000; 1243

Zero block base computation

3002: 0000: 0000: 0028: bcd0: 2347: 800d: 4243

3002: 0: 0: 28: bcd0: 2347: 800d: 4243

3002: : 28: bcd0: 2347: 800d: 4243

3002: 0: 0: 28: bcd0: 2347: 0: 4243

3002: : 28: bcd0: 2347: : 4243
0:0 (or) 0
0 0:0 } in valid

one time computation:

3002: 0: 0: 28: bcd0: 2347: 0: 4243

3002: : 0: 0: 28: bcd0: 2347: : 4243



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Link local IP

fe80:0000:0000:0000:13a6:ba98:97ab:9493
mac

Fe80::

EUI 64

ff:fe

Fe80:: mac add

2a03:2880:f11c:8183:face:b00c:0000:25de

2a03:2880:f351:0022:face:b00c:0000:4420

2001:4860:4860:0000:0000:0000:0000:8888

2001:4860:4860:0000:0235:0316:0021:8844